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Citizens' appeals classification using a mixed method: A case study of Yichang government public service hotline data

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Abstract

The study investigated the automatic classification of citizens' appeals from the "12345" service hotline in a Chinese local government. The approach employed a mixed method with Large Language Models (LLMs) and word embeddings, specifically using ChatGlm2-6b and Moka Massive Mixed Embedding (M3E) respectively. Firstly, taxonomy was developed with Delphi method, and classification categories were determined by experts. The standard element library was built by incorporating weighting factors assigned to elements within each category. Secondly, ChatGlm2-6b was employed to extract the "topics" and "problems" from each new appeal. Next, word embeddings were utilized to compute the similarities between "topics", "problems", and standard elements. Finally, the categories of new appeals were determined with weighted voting. The results indicate that the top five categories of citizens' appeals are "Housing", "Daily life", "Urban management", "Transportation", and "Employment". The accuracy of this method is 0.83. The results can promote early warning systems for emergencies, mining and identification of citizens' appeals, and supervision of social governance.

Keywords

citizens' appeals classification, large language models, word embeddings, local government

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Introduction

Timely responses to citizens' appeals act as a positive feedback mechanism, encouraging increased appeals from citizens. This process opens up channels for effective communication between governments and citizens, thereby addressing information asymmetry, responsiveness, and timeliness issues (Center on Data and Governance Tsinghua University, 2022). Since the inception of the "12345" citizens' hotline in Chinese local governments in 2019, it has significantly enhanced the government's response rate, resolution rate, and satisfaction rate. This hotline operates through a comprehensive set of mechanisms, not only facilitating the handling of incoming calls but also fostering connections within and outside the administrative system. As a result, it effectively mitigates information asymmetry between different levels of administration, promoting civic participation, policy implementation, and the government's responsiveness to societal needs (Ianniello et al., 2019; Zhang, 2022)

Unlike the traditional 'top-down' administrative order distribution, appeals (comprising requests and complaints), gathered through the "12345" hotline service, are manually sorted and directed by hotline operators to respective policy executors. However, governments often fail to respond to them properly; they face many difficulties in properly handling the

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large number of requests received and intending to public concerns. This is because citizens' appeals accepted by public institutions are more likely to be in the form of unstructured data. Therefore, any mechanical processing, such as automatic classification, is bound to be relatively difficult. Usually, the staff in charge of these requests must read the requests manually and categorize them. Due to limited working experience, operators may not consistently grasp the nature of incoming appeals, leading to misclassifications and reduced response efficiency. This is not only cost- and time-consuming, but it also ties up precious human resources away from the more essential governance tasks at hand. Additionally, many appeals involve multiple departments, causing them to cycle between departments, evading responsibility, stalling, or even facing rejection for resolution. Consequently, extensive time is spent liaising between departments to address appeals, significantly slowing down the resolution process. Moreover, there's often information loss during handling, resulting in unresolved issues, repetitive applications from citizens, and therefore, increased citizen dissatisfaction. Thus, many governments are interested in developing automatic classification systems for more efficient workflow. It's imperative to accurately classify citizens' appeals and standardize departmental allocation for each type of request to address these systemic inefficiencies.

Due to the ability of shrinking the gap between human and computer language comprehension, typically, Large Language Models (LLMs) and word embeddings have been used for different text classification tasks dealing with diverse aspects. LLMs showcase strong performance in tasks for which they were not initially designed, through few-shot learning (Brown et al., 2020) or even without specific task-oriented learning (Wei et al., 2021). This adaptability is attributed to these models' capacity to assimilate common-sense knowledge, enabling them to comprehend language, generate text, and engage in reasoning, and modeling (Wu et al., 2023). Utilizing LLMs for the classification of citizens' appeals has proven promising, given their pretraining on extensive collections of unlabeled texts. Furthermore, Word Embeddings has found widespread application in various domains such as emotion analysis (Widmann, 2022), request classification (Kim and Hong, 2021), emergency identification (Devaraj et al., 2020), and satisfaction inference (Flores et al., 2022) among citizens' interactions. Despite this, few studies deeply investigate the capabilities of LLMs and word embeddings to capture the key facts within citizens' appeal text. It is important in local governments, specifically with citizens' appeals, to extract the required and exact information, so an in-depth analysis of these natural language processing (NLP) techniques with application to text classification is becoming an urge.

This article focuses on citizen appeals within the Yichang government and presents the outcomes of employing a system that leverages LLMs and word embeddings for the automated classification of citizens' appeals. To this end, ChatGlm2-6b (Zhipuqingyan, 2023) and Moka Massive Mixed Embedding (M3E) (MokaAI, 2023) have been used as LLMs and word embeddings techniques, respectively to test their abilities to extract, represent and classify citizens' appeals text. To the best of our knowledge, the proposed work represents the first attempt to methodically test the ability of LLMs and word embeddings to text classification in citizens' appeals text. We propose examining these models with 93,971 citizens' appeals collected through the "12345" citizens' hotline of the Yichang government between March 12th and September 12th in 2023. Yichang is a city in the central China area. It's chosen for its data accessibility and representativeness. The Yichang government provides accessible or cooperative data on citizen appeals, making it a viable source for the research. And it is representative in terms of policy implementation and citizen feedback mechanisms of "12345" hotline in the central region.

We list the major contributions of this work as follows: theoretically, it (1) introduces a new mixed method in handling citizen appeal data, contributing to the development of NLP techniques in government data processing; (2) evaluates the accuracy of ChatGlm2-6b and M3E mixed method in large-scale Chinese datasets, providing reference for model improvement; practically, it (1) automates the classification process and determine the appropriate executive body to which the appeals should be assigned, enabling the government to respond and resolve issues more effectively; (2) has the potential to be adapted to other local or central government contexts, which can improve government management and decision-making, especially in resource-limited settings.

Literature review

Automatic citizens' appeals analysis

Analyzing appeals and complaints from citizens plays a vital role in the analysis of civic e-participation (Kabanov, 2022). Researchers emphasized successful

cases of using citizens' appeals in response to emergencies, for example, during the COVID-19 pandemic in China (Chakkarwar and Tamane, 2020; Zhang et al., 2022). Lu et al. (2023) measured the textual complexity based on the textual data of the 1.15 million online messages left by the citizens to government leaders on China's online public service platform and explored its impact on government responses. Markovets et al. (2019) analyzed citizens' appeals in heterogeneous web services and proposed a formal model to determine the level of trust in the author, such as monitoring, attestation, organization, and personification.

NLP methods, as one of the common automatic text analytics tools, are widely used in the analysis of citizens' appeals (Feng et al., 2023). The NLP methods can serve as a text classifier based on the pre-trained language model and natural entity recognition combined with approximate string matching can be used to identify approximate locations of public events in messages (Antonov et al., 2023). A convolutional neural network also can automate the process of distributing citizens into categories with the help of the pre-trained model of Natural Language BERT and sciBERT (Begen and Chugunov, 2019; Bunova and Serova, 2022). Word embeddings are also used in the automatic classification of citizens' appeals. For instance, Mangusheva et al. (2021) calculated the distance vector from each new document to other documents respectively and constructed a distance matrix using word2vec and Word Mover's Distance algorithms. Kim and Hong (2021) proposed an automatic classification model for unstructured data by using transportation-related citizen requests as an example. A combination of unsupervised and supervised learning was applied to the data and word embeddings were used to improve the quality of textual data.

LLMs and word embeddings for text classification

Currently, there are many methods most often used to classify text data, such as manual method, using developed rules, machine learning, etc. As the appeals from citizens are essentially a large array of textual data, often poorly structured (Göbel and Li, 2021). In this case, the manual way and the usage of rules can only be suitable to be applied locally to search for specific words, instead of processing a large array of data. As for machine learning methods, latent dirichlet allocation (LDA), vector space model (VSM), artificial neural networks (ANN), K-nearest neighbor (KNN), Naives Bayes (NB) and support vector machine (SVM) have been applied in automatic text classification.

The word embeddings method is a type of VSM. It's an unsupervised technique used to map words or phrases from a text to a corresponding vector of real numbers. The obtained vector space through embedding models preserves the contextual similarity of words. Some mixed methods have been examined for better performance in text classification, such as LDA combined with VSM (Carrera-Trejo et al., 2015), VSM or word embeddings combined with ANN (Rajan et al., 2009; Jiang et al., 2016), etc. In these research, it's shown that word embeddings and related methods can reduce the feature size and improve text classification accuracy by integrating rich document representations.

With the birth of LLMs, this approach is currently the most promising one (Bunova and Serova, 2022). By comparing the text classification results of LLMs and other well-established text transformation techniques, including word embeddings, Gidrol (2023) found while LLMs are promising, the essence of obtaining valuable outcomes is in feeding them with the right context and sculpting their responses effectively. This comparison not only highlights the potential of LLMs but also underscores the nuances of engaging meaningfully with these advanced models. LLMs can also be used similarly to earlier word embeddings, as new texts can be converted into LLM-specific vectorized representations, which can be input as features in standard classifiers (Bonikowski et al., 2022; Rodriguez and Spirling, 2022). Notably, Chae and Davidson's (2023) results show that LLMs can accurately identify stances in social media texts, outperforming conventional machine learning approaches using word embedding representations.

However, it's not clear the accuracy of LLMs and word embeddings leveraged methods to perform complex text classification tasks compared with conventional baselines. Besides, the implications of LLMs and word embeddings for public sectors have not been underscored so far. Previous studies about LLMs and governments focus only on establishing chatbox (Cardoso, 2023), open government data (Loukis et al., 2023), and public acceptance (Yang and Wang, 2023). There has been no research about the capabilities it can provide to citizens' appeals combined with word embeddings, which is the gap that this study aims to fill.

Research design

Research method

In this study, citizens' appeals are classified mainly based on LLMs and word embeddings. The main workflow involves two parts, appeals taxonomy development and appeals classifier construction, as shown in Figure 1.

In the appeals taxonomy development part, firstly, the Delphi method is used. The taxonomy, which contains 20 main categories and 131 subcategories, is initially proposed based on experts' opinions, civil servants' working experience, and existing classification standards. Then, pre-labeled appeals are selected in each category and standard elements of them are extracted by experts. The weighting factors of standard elements paired with each category are calculated. A knowledge graph of the standard element library is built to reveal categories, subcategories, standard elements, and their relations.

As for appeals classifier construction, firstly, LLMs are applied to comprehend and extract each appeal's "topic" and "problem". Secondly, word embeddings are used to calculate the similarity matrix of topics, problems, and standard elements. The standard elements with high similarities are then chosen and the weighting factors in each category are added up to determine the subcategory and category of each appeal. The last stage of the process is dispatching to the intended performer to whom the appeals should be addressed.

Delphi method. The taxonomy development process, based on Nickerson et al. (2013), comprises four process steps: (1) choose a meta-characteristic of the object of interest, (2) specify dimensions, (3) define necessary conditions for the taxonomy, and (4) conceptualize characteristics. Our meta-characteristic comprises the design and application of Delphi methods. The specification of the dimensions of the object of interest is addressed with the proposal of categories and subcategories. The necessary conditions for the taxonomy are that it is concise, robust, comprehensive, extendible, and explanatory in the field of appeals. Conceptualizing the characteristics is to build a standard elements library in this study. These standard elements are standardized topics of appeals selected by experts.

Two rounds Delphi method are used to develop the taxonomy. The first round is to determine categories and subcategories and the second round is to define standard elements library. The experts at the top of their field of knowledge and are interested in a wide range of matters in their field are required (Strasser, 2017). As for local governments, some aspects of working experience and knowledge remain tacit in the minds of civil servants, administrators, or experts. So, the 33 executives of government departments are selected. Firstly, a face-to-face meeting is organized by the coordinator to keep the experts informed of the aim and procedures of the research. Secondly, the first-round questionnaire is distributed to ask them to generate the main problems of citizens' appeals during their work. 200 representative appeals in each main category are selected and submitted by the experts. Finally, the second-round questionnaire is conducted to ask them to identify the standard elements of the appeals.

LLMs. ChatGlm2-6b is chosen to conduct text extraction. ChatGLM2-6B is an open bilingual language model that uses technology similar to ChatGPT,

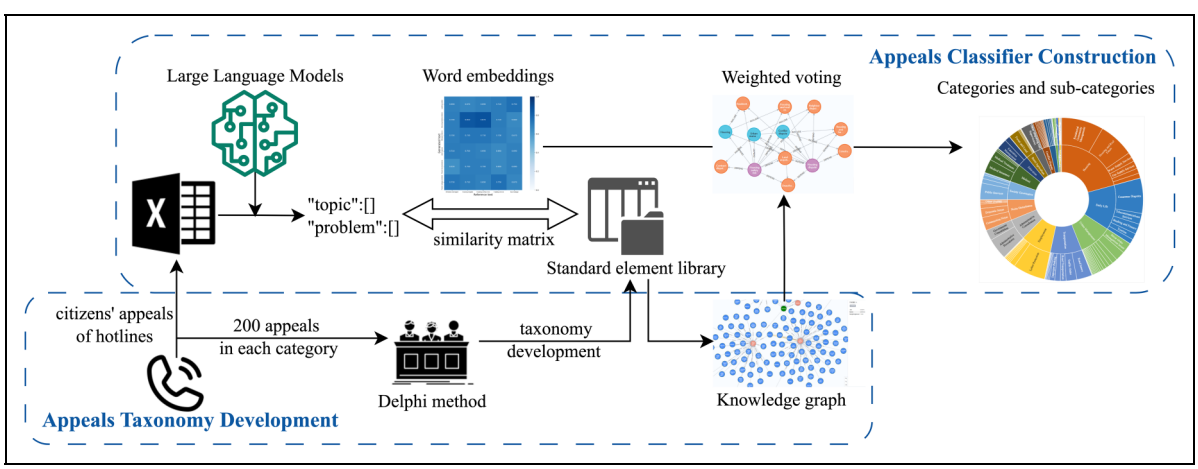

Figure 1. The research methods and processes.

optimized for Chinese question answering and dialogue. ChatGLM2-6B has undergone pre-training with 1.4 T bilingual tokens and human preference alignment training. It's shown that the number of parameters of ChatGlm2-6b is sufficient for citizens' appeals analysis. Besides, longer context and more efficient inference of ChatGLM2-6B allow for more rounds of dialogue, more efficient inference speed, and lower GPU memory usage.

To avoid AI "hallucinations" and ensure data quality and credibility, several strategies are taken. First, Al hallucinations might occur when there are no corresponding and relevant texts in the existing LLM training data. Previous messages influence the model's context and response, enabling the user to reference earlier messages or elucidate statements, akin to standard chat interactions. Besides, the model temperature is set to zero to minimize the randomness of the answer. The last strategy is prompt tuning. The prompt includes text contents, requests, and output requirements. As for requests, clear and direct user requests yield more precise responses.

Word embeddings. The goal of word embedding is mapping the words in unlabeled text data to a continuously valued low dimensional space, in order to capture the internal semantic and syntactic information (Li and Yang, 2018). It can resolve the task of determining semantic similarities between texts (Kenter and De Rijke, 2015). With word embeddings models, the algorithm maps a vector to each word. Close words correspond to close vectors. A measure of the proximity of words is their contextual similarity: close words are found in the text next to the same words. As machines require numerical inputs to perform computations, text embeddings are a crucial component of many downstream NLP applications.

M3E Model is used to conduct text representation in this research. This model is trained, open-sourced, and evaluated by MokaAI, using uniem for the training script and Massive Text Embedding Benchmark(zh) for the evaluation benchmark (Muennighoff, 2022). This model is trained on a dataset of 22 millons Chinese sentences and supports bilingual homogeneous text similarity calculation and heterogeneous text retrieval. It has 110 M parameters and 768 dimensions and converts natural language into dense vectors with an accuracy of 0.6157 in text classification tasks, behaving better than text2vec and openai text-embedding-ada-002 (MokaAI, 2023).

Considering the high accuracy in large-scale sentence pair datasets of Chinese corpora, M3E is chosen to realize text classification of citizens' appeals by obtaining vector representations of words.

Data collection and cleaning

Data collection

Data of citizens' appeals to the "12345" hotline of the Yichang government are used. 93,971 citizens' appeals between March 12th and September 12th in 2023 are collected. They are transformed from hotline calls into texts and disclosed on the official website of "12345" citizens' hotlines of the local government (http://www.yichang.gov.cn/Szrx/lists2?areaid=1&t=1). Yichang City, situated in the central China area, is chosen as it has developed this service for 4 years. Citizens' hotlines have become the main channel for citizens to give feedback on their appeals.

Data cleaning

To improve the accuracy and performance of the model, data cleaning is conducted in the experiment. Firstly, re Python regex package is used to remove punctuation signs, hashtags, duplicate spaces, and data without value. It is also applied to data anonymization to eliminate sensitive and private information, i.e., ID number, mobile phone number, and name. The data examination revealed that the maximum message length is 239, the minimum is 12 and the average is 59.23. The dataset structure includes submitted appeals with their attributes. This parameter set contains the fields: appeal_category, appeal_sub_category, submitted_time, title, location (district), content, appeal_state.

Appeals taxonomy development

Categories and subcategories

Categories and subcategories are proposed mainly based on the Delphi method. Besides, these experts also refer to Government Service Hotline Request Classification Standards of local governments in other districts, such as Hainan Province, Guangdong Province, etc. The categories are adjusted according to the essential and urgent issues that are mentioned most frequently by local citizens. Based on this, the proposed taxonomy

includes 20 main categories and 131 subcategories of citizens' appeals. The main categories include Daily life, Food, Transportation, Employment, Education, Medical, Housing, Mental health, Urban management, Rural agriculture, Society governance, Housing provident fund, Epidemic prevention and control, Social insurance, Noise disturbance, Conflict dispute, Phone number Consultation, Reporting and supervision, Advice, Administrative consultation. These categories are identified according to the concerns of citizens, the functions of local governments, and civil servants' working experience.

Standard elements library

Each appeal might contain one or more elements. The elements of appeals are extracted and checked manually by experts to be added to the standard element library. We select 200 appeals in each category, except that 2 categories are excluded with no more than 200 appeals. About 5800 standard elements are identified after removing duplicates. Each standard element may belong to several categories and subcategories and has different weighting factors for these categories. The weighting factor (WF_{ij}) of one standard element (SE_i) in each category (C_j) can be calculated as:

$$WF_{ij} = (\text{the frequency of } SE_i)/(\text{the total number of citizen appeals in } C_j)$$

The standard elements, weighting factors, and categories are stored in the knowledge graph which forms the basis for further model training. Neo4j is applied to visualize the standard element library which contains the relationship between standard elements and categories. The weighting factor is specified as the relationship property. As is shown in Figure 2, "Housing" is the main category, and "Rural water supply problem" and "Gas service problem" are subcategories. Taking the standard element "Unreasonable charge" as an example, it belongs to two subcategories. There are 50 citizens' appeals in "Gas service problem" and the frequency of "Unreasonable charge" is 18 so the weighting factor of "Unreasonable charge" in the specific "Gas service problem" subcategory is 0.360.

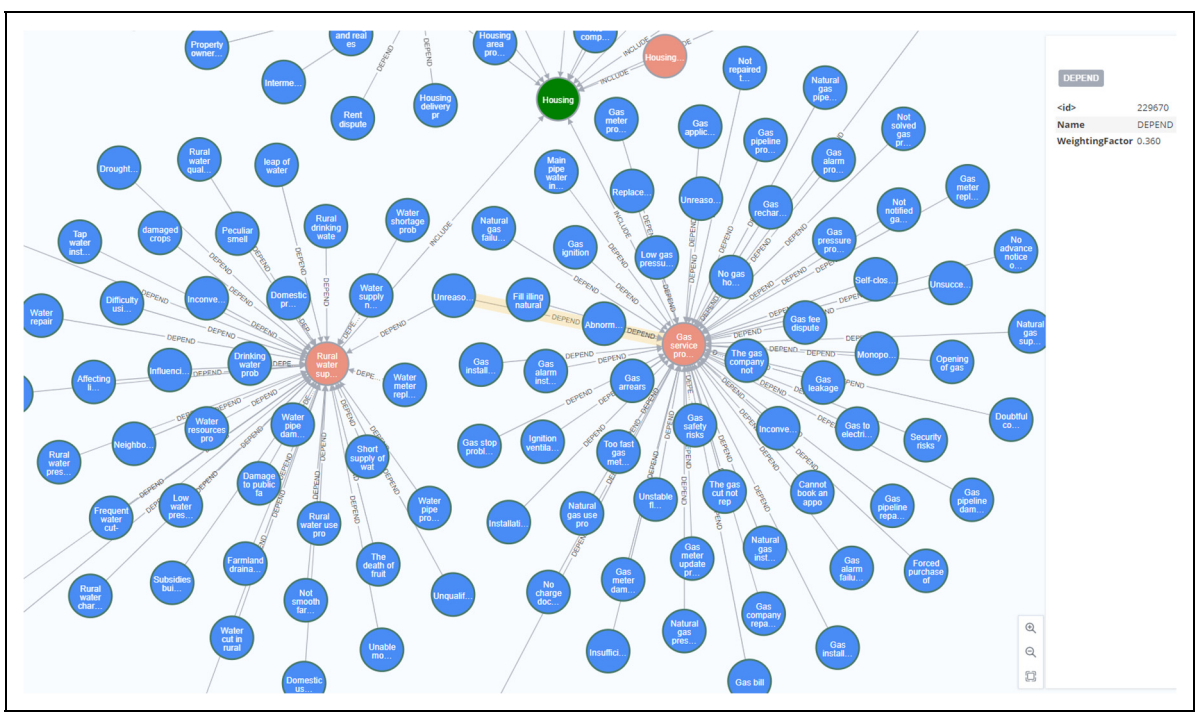

Figure 2. The knowledge graph of the standard element library.

Appeals classifier construction

Text comprehension

The text comprehension task is to extract and summarize the key facts and main idea of the citizens' appeals. The ChatGlm2-6b is employed for text comprehension of citizens' appeals. To avoid AI "hallucinations", examples of text comprehension tasks whose input and output are defined as a reference can be provided to enhance response quality. ChatGlm2-6b was requested to extract the "topics" and "problems" of each appeal. The prompt is "Please summarize the key topics and main problems from the above text. The results are in JSON format. The key field only contains the summarized key topics and main problems. Do not include other fields." "Topics" and "problems" are extracted to ensure a complete understanding of appeals. responses are specified in string format to reduce the randomness and make it easier to filter out meaningless answers. Output requirements define that the results should be output into a JSON format. The extracted "topics" and "problems" vary from 3 to 10 in each appeal and the average is 4.869 (Figure 3).

Text representation

The purpose of text representation is to find the closer standard elements that are similar in meaning with extracted "topics" and "problems". With word embeddings model M3E, the "topics" and "problems" of each appeal are multiplied with vectors of standard elements to calculate the cosine similarity. As is shown in Figure 4, the results of the vectorization operation in parts of elements form the similarity matrix. The generated text is the "topics" and "problems" of each new appeal, and the reference text is standard elements. For example, the elements "Housing rental problem," "Windows damaged," and "Housing dispute" are distributed in 768-dimensional vector space. The cosine similarity between the words "Housing rental problem" and "Housing dispute" (0.850) is more distinct than between "Housing rental problem," and "Windows damaged" (0.700). So, as for generated text "Housing rental problem", its meaning is closer to the standard element "Housing dispute". The result of the standard element will be used to classify the corresponding appeals.

Text classification

Text classification is conducted combined with the knowledge graph of the standard element library and contextual similarities of elements. The result of word embeddings is integrated to map "topics" and "problems" into standard elements. The standard elements with the top 5 highest-ranked similarities that are more than 0.8 between generated texts and them are chosen. In Figure 4, as for the "Housing rental problem" in related new appeals, standard elements "Housing dispute" and "Housing safety risks" would be chosen.

The weighted voting strategy is applied. With the constructed knowledge graph, the weighting factors of the chosen standard elements in the same category are added up. The category with the highest score is determined as the category of the new appeal. After "Housing dispute" and "Housing safety risks"

```
#One of examples for text comprehension tasks
msg1 = ["""
#text contents
"6月18日晚上21:20分左右我在夷陵区东湖大道雅斯超市，购买了一根伊利升级玉米香
雪糕，65g标价2元，但实际收费2.5元，使用微信支付。请处理。
'''At 9:20 pm on 18th. June, I bought an ice cream at Yasi Supermarket, Donghu Avenue,
Yiling District.  The price was labeled as $2, but the actual charge was $2.5. I used WeChat
payment. Please deal with it. '''

#output requirements
Please summarize the key topics and main problems from the above text. The results are in
JSON format. The key field only contains the summarized key topics and main problems. Do
not include other fields." "Topics" and "problems" are extracted to ensure a complete
understanding of appeals.
Output example: {"topic": [], "problem": []}
"""
,
#output contents
"""{
    "topic": [ "消费纠纷", "一般食品消费问题" ] ,
  "problem": [ "雅斯超市消费问题" ]
  #"topic": [ "Consumer Disputes", " General Food Consumption Issues"],
  #"problem": [ "Consumption Issues of Yasi Supermarket"]
} """
]
```

```
#Request ChatGlm2-6b extract the "topic" and "problem" of each appeal
template = """
"{texts}"
Please summarize the key topics and main problems from the above text. The results are in
JSON format. The key field only contains the summarized key topics and main problems. Do
not include other fields." The "topics" and "problems" are extracted to ensure a complete
understanding of appeals.
"""
example = """output example：\n{"topic": [], "problem":[]}"""
openai.api_key = 'none'
openai.api_base = 'http://10.18.43.14:8000/v1'
def get_completion(prompt, model="Chatglm2-6B"):
    prompt = template.format(texts=prompt) + example
    print(prompt)
    messages = [{"role": "user", "content": msg1[0]},
                {"role": "assistant", "content": msg1[1]},
                {"role": "user", "content": msg1[2]},
                {"role": "assistant", "content": msg1[3]},
                {"role": "user", "content": prompt}]
    response = openai.ChatCompletion.create(
        model=model,
        messages=messages,
        temperature=0)
    rst = response.choices[0].message["content"]
    print(rst)
    return rst
```

Figure 3. The prompt of text comprehension task.

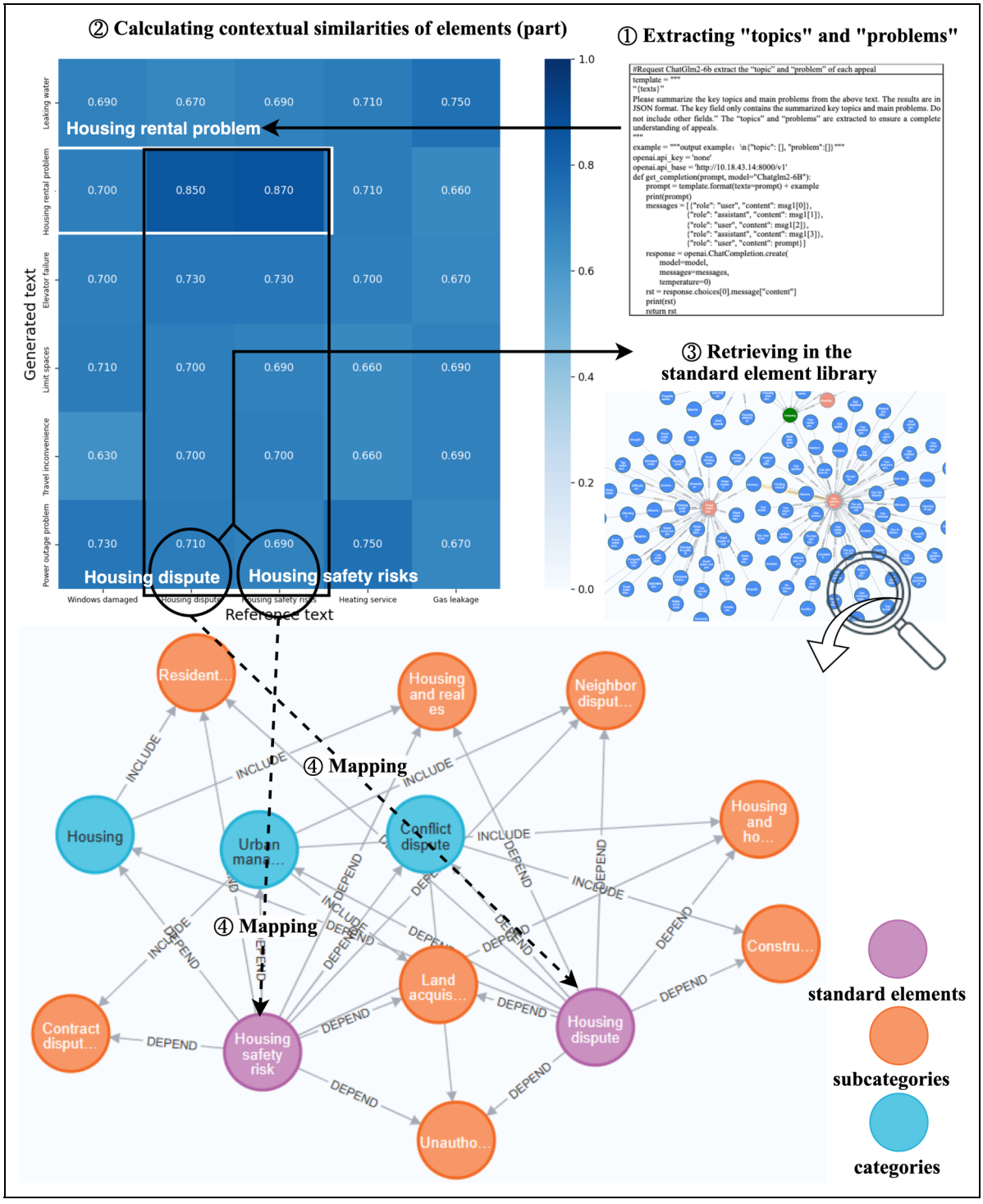

Figure 4. Mapping selected standard elements in the knowledge graph.

are selected as the standard elements with the largest similarity, their weighting factors in subcategories and categories can be retrieved in the knowledge graph of the standard element library in Figure 4. It's shown that they both exist in 6 subcategories: "Residential community management", "Housing and real estate", "Unauthorized construction", "Neighbour disputes", "Housing and homestead disputes", "Land acquisition and demolition disputes". Meanwhile, "Housing dispute" is related to "Construction management issues" and "Housing safety risks" is related to "Contract disputes". As for the main categories,

Table 1. The weighted voting of standard elements.

Weighting factors of standard elements				
Categories and subcategories		Housing dispute	Housing safety risks	Total
Housing	Residential community management	0.86	0.45	**1.31**
	Housing and real estate	0.19	0.21	0.40
Conflict dispute	Construction management issues	0.39	\	0.39
	Unauthorized construction	0.43	0.49	0.92
Urban management	Neighbour disputes	0.02	0.13	0.15
	Housing and homestead disputes	0.89	0.32	1.21
	Land acquisition and demolition disputes	0.18	0.38	0.46
	Contract disputes	\	0.49	0.49

Table 2. The result of main categories.

Main category	Number of citizens' appeals	Percentage (%)	Main category	Number of citizens' appeals	Percentage (%)
Housing	19,542	20.796	Rural agriculture	2575	2.740
Daily life	12,554	13.359	Reporting and supervision	2350	2.501
Urban management	9203	9.793	Conflict dispute	2011	2.140
Transportation	8867	9.436	Social insurance	1719	1.829
Employment	8535	9.083	Food	649	0.691
Administrative consultation	5943	6.324	Phone number consultation	439	0.467
Noise disturbance	5772	6.142	Epidemic prevention and control	397	0.422
Society governance	5101	5.428	Housing provident fund	242	0.258
Medical	5064	5.389	Advice	4	0.004
Education	3001	3.194	Mental health	3	0.003

they both belong to "Housing", "Conflict dispute" and "Urban management." The weighting factors of "Housing dispute" and "Housing safety risks" paired with each subcategory and category are added up, as shown in Table 1. The weighting factor of "Housing dispute" and "Housing safety risks" in "Residential community management" is 0.86 and 0.45 respectively whose sum is the highest score (1.31) so "Residential community management" is voted as the subcategory of the new appeal. And "Housing" is determined as its category.

Results

Classification results and applications

Through the experiment of classification, 93,971 citizens' appeals between March 12th and September 12th in 2023, are divided into 20 main categories and 131 subcategories. The results of the main categories and subcategories are shown in Table 2 and Figure 5. "Housing" is the citizens' most concerned issue whose number of citizens' appeals is 19,542 and the proportion is 20.796%. Its most frequently mentioned subcategories are "Residential community management", "Housing and real estate", "Water supply service" and "Gas supply service". Citizens care about issues of "Daily life" too, with 12,554 appeals (13.359%), ranking second in all categories. The number of citizens' appeals in "Urban management" is 9203 and represents 9.793%. Citizens' appeals about "Transportation" and "Employment" are 8867(9.436%) and 8535(9.083%) respectively.

With this classifier, the key facts of citizens' appeals can be identified and dispatched automatically and efficiently, decreasing time, effort, and misclassification errors involved in manual classification. It can also promote early warning of emergencies, citizens' appeals mining and identification, and social governance supervision.

Firstly, the proposed method helps to quickly obtain structured information about emergent problems that need to be addressed. For example, urban governors

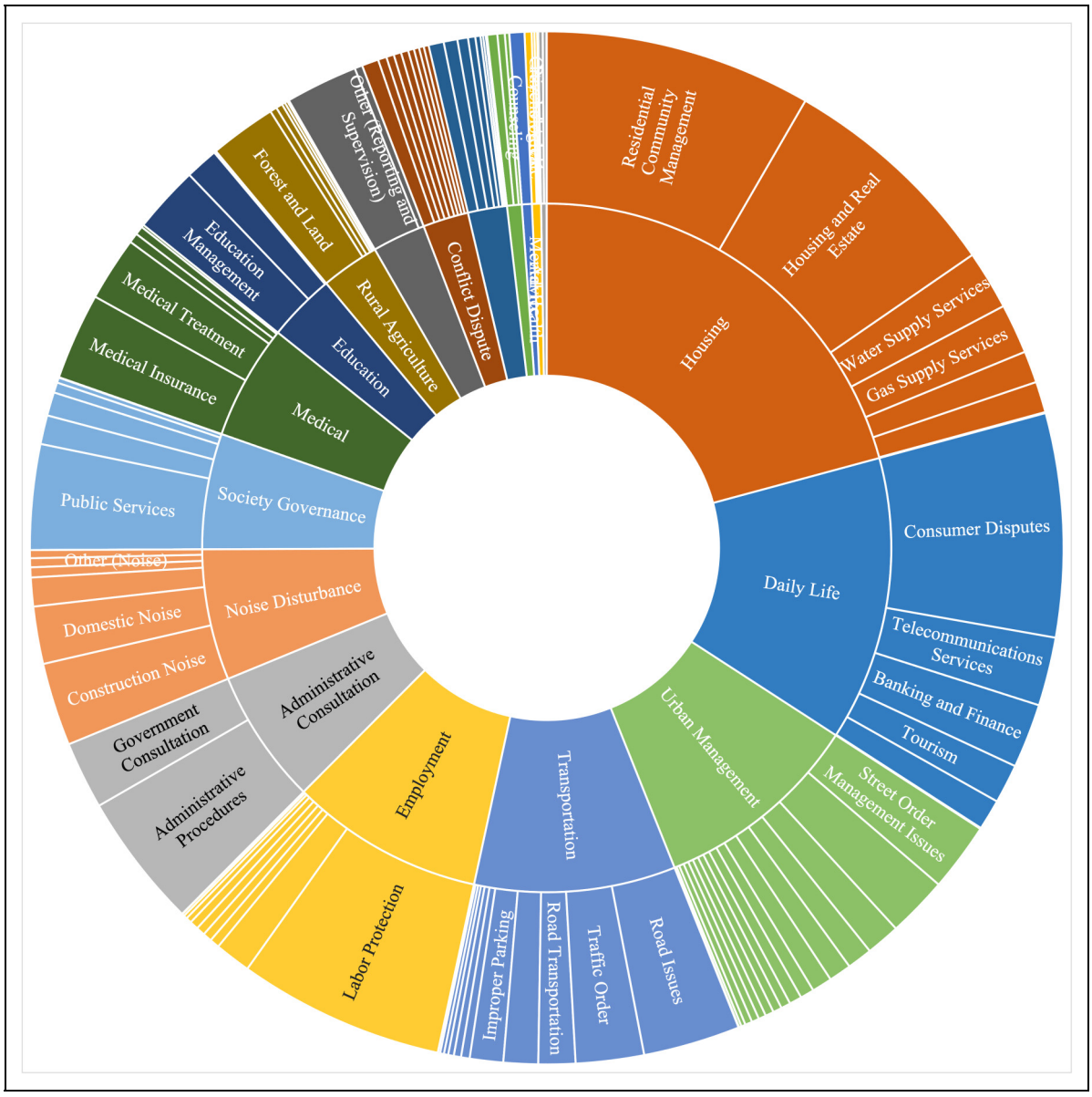

Figure 5. The results of categories and subcategories.

can request LLMs to identify urgent appeals containing "water outage", "power outage" and "gas outage" and emergent situations about "commit suicide", "jumping from a building", "physical threats", "violence", "conflict", "calling the police", "criminal cases", etc. So, the related departments can focus on these special issues and take action immediately to provide support for social stability.

Secondly, LLMs can further mine and extract entities of these appeals to help the government analyze the appeals from diverse dimensions such as the temporal and spatial distribution of citizens' appeals. Entities like places can also be extracted by the LLM and visualized on the map with precise matching of geographic location, as is shown in Figure 6. For instance, between September 5th and September 12th in 2023, over 300 appeals have been received and the distribution of them centers on the several central business districts along with the river where the city develops best and has the largest population. Meanwhile, in economically prosperous districts, channels for political participation are widely disseminated, enhancing the willingness and ability of citizens to participate in social governance, which drives citizens to express their appeals actively.

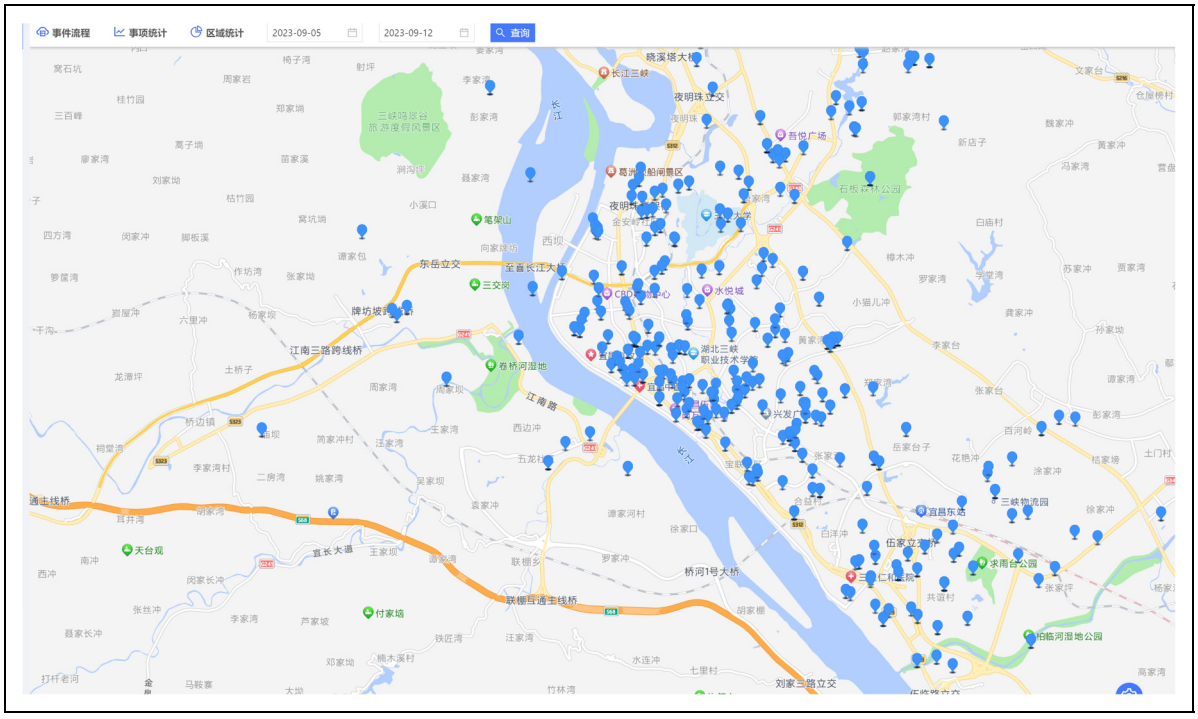

Figure 6. Map of Yichang city with citizens' appeals from September 5th –12th.

Thirdly, the method can be applied to assess the quality and performance of city government services. With LLMs, "secondary processing" appeals which are divided into "No satisfactory response", "Not processed" and "Refused to be processed" can easily be recognized. These cases will be specifically refocused to improve the satisfaction of citizens. In the meanwhile, unsatisfactory feedback will be considered in the performance evaluation system of civil servants.

Model Effect Evaluation

This research compares and analyses several solutions' performance: manual method, "classification by LLMs" method, "extraction and classification by LLMs" and "extraction by word embeddings and classification by LLMs" method respectively. These methods' accuracy and response time are tested and compared to the manual method. The 3600 appeals selected randomly (180 in each main category) are used as a testing set. To simplify the evaluation process, only the results of the main categories are examined.

1. The manual method is conducted by experts.
2. The "classification by LLMs" method is to request ChatGlm2-6b to directly classify the appeals. The prompt is "These appeals can be divided into 20 categories which are Housing, Daily life, Urban management … Please classify the appeal and output the result. The results are in JSON format."
3. The "extraction and classification by LLMs" method is to request ChatGlm2-6b to extract "topics" and "problems" and then classify the appeals with the results. The prompt is "Please summarize the key topics and main problems from the above text. The generated topics and problems can be divided into 20 categories which are Housing, Daily life, Urban management … Please classify the generated topics and problems and output the result. The results are in JSON format."
4. The last method is the mixed method adopted in this study.

Since the appeals are classified by experts in manual method, the accuracy of it can be seen as 1 but it takes about 5 min on average to figure out one appeal. The methods that only use LLMs are faster but with low accuracy which is 0.57 and 0.68 respectively. It echoes Sun et al. 's (2023) findings that the performances of LLMs still significantly underperform fine-tuned models in the task of text

Table 3. The model effect evaluation.

Evaluation index Research method	Number of correct classifications	Accuracy	Average response time
Manual method	3600	1	About 5 min
Classification by LLMs	2052	0.57	1.14 s
Extraction and classification by LLMs	2447	0.68	2.32 s
Extraction by LLMs and classification by word embeddings	2988	0.83	3.46 s

classification. This is due to (1) the lack of reasoning ability in addressing complex linguistic phenomena; and (2) the limited number of tokens allowed in in-context learning. In comparison, the method we use in this study shows much higher accuracy (0.83). As shown in Table 3, the combination with word embeddings improves the performance in text classification.

Discussion and conclusion

Investigating the automatic classification of citizens' appeals enables governments to respond to their citizens more swiftly and to better utilize their human resources. With the proposed LLMs and word embeddings method, this study identified and classified more than 93,000 citizens' appeals to a Chinese local government. Firstly, the taxonomy is developed with the Delphi method. The categories and subcategories are identified. The standard element library is constructed in a knowledge graph with standard elements and their weighting factors in categories. Secondly, ChatGlm2-6b is selected to conduct text comprehension tasks and extract topics and problems of appeals. Next, M3E is used to carry out text representation tasks and calculate the similarity matrix of elements. Finally, the appeals are classified with weighted voting with factors paired with standard elements and categories. The results show that most citizens are concerned with "Housing", "Daily life", "Urban management", "Transportation", and "Employment". The accuracy of this method is 0.83.

Unlike previous automatic classification studies conducted using relatively high-quality data such as academic materials and media articles (Fang et al., 2020; Gargiulo et al., 2019), the presented study processes the classification of a large number of citizens' appeals extracted from unstructured text. Specifically, the dataset used in this study is differentiated in the following aspects: (1) a larger number of categories are present, (2) multiple and imbalanced categories, and (3) including many typos, neologisms, and abbreviations directly written by citizens. To this end, first, the categories that formed the basis of the classification of the citizen requests are confirmed and consequently optimized by experts. The standard element library is built to describe the coverage and characteristics of categories. Second, ChatGlm2-6b continuously shows the best outcome and is selected to accurately comprehend citizen's appeals. It's shown that the model can handle processing low-quality Chinese text data on specific topics. Third, word embeddings are a key step to obtaining desirable outcomes. Citizens' appeals data contain informal language that is of a quality lower than that of news or academic documents, and this in turn makes the task of processing difficult. In this study, the use of M3E calculates the similarity of key facts of appeals with standard elements to determine their categories to reduce ambiguity and normalize textual data.

This study makes the following contributions. It's the first time that LLMs combine with word embeddings to analyze citizens' appeals. The mixed method for automatic text classification proposed by our study can be extended for the identification of such needs and fast response. This approach is expected to be used in the processing and classification of other citizen requests and low-quality text data on specific topics in the future. The results of the study open up prospects for the application and development of AI-assisted systems for public policies. Within these systems, demand recognition and risk detection are possible through the discovery and identification of citizens' appeals. Policies can be made according to the dynamically updated and fine-grained requests from citizens, thus improving government management and service delivery. On the other hand, the reduced time and workloads will enable municipal employees to concentrate more on core tasks. A more accurate classification will provide the government with objective criteria

for declaring clear rights and responsibilities, reasonable division of labor, and collaboration. In this sense, our work can be recognized as another piece of evidence that proves diverse deep learning-related theories by applying them to various unstructured data of citizen requests.

This study suffers from certain limitations. As a further stage, it is planned to train the classifier of LLMs in more specific domains. The requests proposed to LLMs could be more accurate and more entities can be extracted for further analysis.

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References

Antonov A, Vidiasova L and Chugunov A (2023) Detecting Public Spaces and Possibilities of Risk Situations in Them via Social Media Data. In: Coman A and Vasilache S (eds) *Social Computing and Social Media*. HCII 2023. Lecture Notes in Computer Science, vol 14025, Copenhagen, Denmark, 23–28 July 2023. Cham: Springer, 3–13. DOI: 10.1007/978-3-031-35915-6_1.

Begen PN and Chugunov AV (2019) Intellectual classifier development of citizens' messages on the «Our St. Petersburg» portal: Experience in using machine learning methods. *Scientific Services & Internet* 21: 131–140.

Bonikowski B, Luo Y and Stuhler O (2022) Politics as usual? Measuring populism, nationalism, and authoritarianism in US presidential campaigns (1952–2020) with neural language models. *Sociological Methods & Research* 51(4): 1721–1787.

Brown T, Mann B, Ryder N, et al. (2020) Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33: 1877–1901.

Bunova EV and Serova VS (2022) Methodology for solving problems of classification of appeals/requests of citizens to the "hotline" of the president of the Russian Federation. *Вестник Южно-Уральского Государственного Университета. Серия: Компьютерные Технологии, Управление, Радиоэлектроника* 22(2): 29–40.

Cardoso AG (2023) Do we need a Chat-GPT-Gov? The importance of technology for effective access to public information. *The Importance of Technology for Effective Access to Public Information. (January 7, 2023)*.

Carrera-Trejo JV, Sidorov G, Miranda-Jiménez S, et al. (2015) Latent Dirichlet Allocation complement in the vector space model for Multi-Label Text Classification. *International Journal of Combinatorial Optimization Problems and Informatics* 6(1): 7–19.

Center on Data and Governance Tsinghua University (2022) *Development Report of Digitalization and Intellectualization for Government Hotline*.

Chae Y and Davidson T (2023) Large Language Models for Text Classification: From Zero-shot Learning to Fine-Tuning. *Open Science Foundation*, 21–39. DOI: 10.31235/osf.io/sthwk.

Chakkarwar V and Tamane S (2020) Social media analytics during pandemic for Covid19 using topic modeling. In: 2020 International Conference on Smart Innovations in Design, Environment, Management, Planning and Computing (ICSIDEMPC), Aurangabad, India, 30–31 October 2020, pp.279–282. DOI: 10.1109/ICSIDEMPC49020.2020.9299617.

Devaraj A, Murthy D and Dontula A (2020) Machine-learning methods for identifying social media-based requests for urgent help during hurricanes. *International Journal of Disaster Risk Reduction* 51: 1–12. DOI: 10.1016/j.ijdrr.2020.101757.

Fang W, Luo H, Xu S, et al. (2020) Automated text classification of near-misses from safety reports: An improved deep learning approach. *Advanced Engineering Informatics* 44: 1–8. DOI: 10.1016/j.aei.2020.101060.

Feng X, Wang C and Wang J (2023) Understanding how the expression of online citizen petitions influences the government responses in China: An empirical study with automatic text analytics. *Information Processing & Management* 60(3): 1–21. DOI: 10.1016/j.ipm.2023.103330.

Flores AM, Pavan MC and Paraboni I (2022) User profiling and satisfaction inference in public information access services. *Journal of Intelligent Information Systems* 58: 67–89.

Gargiulo F, Silvestri S, Ciampi M, et al. (2019) Deep neural network for hierarchical extreme multi-label text classification. *Applied Soft Computing* 79: 125–138.

Gidrol J-B (2023) Text classification with LLMs. Available at: https://jbgidrol.com/paper/Text_classification_with_LLMs.pdf (accessed 20 June 2024).

Göbel C and Li J (2021) From bulletin boards to big data: The origins and evolution of public complaint websites in China. *Journal of Current Chinese Affairs* 50(1): 39–62.

Ianniello M, Iacuzzi S, Fedele P, et al. (2019) Obstacles and solutions on the ladder of citizen participation: A systematic review. *Public Management Review* 21(1): 21–46.

Jiang S, Lewris J, Voltmer M, et al. (2016) Integrating rich document representations for text classification. In: 2016 IEEE Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 29-29 April 2016, pp.303–308. IEEE.

Kabanov Y (2022) Refining the UN E-participation Index: Introducing the deliberative assessment using the Varieties of Democracy data. *Government Information Quarterly* 39(1): 1–15. DOI: 10.1016/j.giq.2021.101656.

Kenter T and De Rijke M (2015) Short text similarity with word embeddings. In: Proceedings of the 24th ACM International Conference on Information and Knowledge Management, Melbourne, Australia, 18–23 October 2015, pp.1411–1420. New York: Association for Computing Machinery.

Kim N and Hong S (2021) Automatic classification of citizen requests for transportation using deep learning: Case study from Boston city. *Information Processing & Management* 58(1): 1–13. DOI: 10.1016/j.ipm.2020.102410.

Li Y and Yang T (2018) Word embedding for understanding natural language: A survey. *Guide to Big Data Applications* 26: 83–104. DOI: 10.1007/978-3-319-53817-4_4.

Loukis E, Saxena S, Rizun N, et al. (2023) ChatGPT application vis-a-vis open government data (OGD): Capabilities, public values, issues and a research agenda. In: International Conference on Electronic Government, Budapest, Hungary, 5–7 September 2023, pp.95–110. Springer, Cham.

Lu L, Xu J and Wei J (2023) Understanding the effects of the textual complexity on government communication: Insights from China's online public service platform. *Telematics and Informatics* 83: 1–17. DOI: 10.1016/j.tele.2023.102028.

Mangusheva A, Kvaratskhelia A, Rakhimov D, et al. (2021) Automatic classification of Citizens' appeals. In: 2021 International Conference on Information Technology and Nanotechnology (ITNT), Samara, Russian Federation, 2021, pp.1–4. DOI: 10.1109/ITNT52450.2021.9649330.

Markovets O, Pazderska R, Dumanskyi N, et al. (2019) Analysis of citizens' appeals in heterogeneous web services. In: CEUR Workshop Proceedings, Lviv, Ukraine, 16-17 May 2019, pp.184–198. Available at: https://ceur-ws.org/Vol-2392/paper14.pdf (accessed 20 June 2024).

MokaAI (2023) Moka massive mixed embedding. Available at: https://huggingface.co/moka-ai/m3e-base (accessed 20 June 2024).

Muennighoff N (2022) MTEB: Massive Text Embedding Benchmark. Available at: https://huggingface.co/blog/mteb (accessed 20 June 2024).

Nickerson RC, Varshney U and Muntermann J (2013) A method for taxonomy development and its application in information systems. *European Journal of Information Systems* 22(3): 336–359.

Rajan K, Ramalingam V, Ganesan M, et al. (2009) Automatic classification of Tamil documents using vector space model and artificial neural network. *Expert Systems with Applications* 36(8): 10914–10918.

Rodriguez PL and Spirling A (2022) Word embeddings: What works, what doesn't, and how to tell the difference for applied research. *The Journal of Politics* 84(1): 101–115.

Strasser A (2017) Delphi method variants in information systems research: Taxonomy development and application. *Electronic Journal of Business Research Methods* 15(2): 120–133.

Sun X, Li X, Li J, et al. (2023) Text Classification via Large Language Models. *ArXiv Preprint ArXiv:2305.08377*. DOI: 10.48550/arXiv.2305.08377.

Wei J, Bosma M, Zhao VY, et al. (2021) Finetuned language models are zero-shot learners. *ArXiv Preprint ArXiv:2109.01652*. DOI: 10.48550/arXiv.2109.01652.

Widmann T (2022) Fear, hope, and COVID-19: Emotional elite rhetoric and its impact on the public during the first wave of the COVID-19 pandemic. *Political Psychology* 43(5): 827–850.

Wu T, He S, Liu J, et al. (2023) A brief overview of ChatGPT: The history, status quo and potential future development. *IEEE/CAA Journal of Automatica Sinica* 10(5): 1122–1136.

Yang L and Wang J (2023) Factors influencing initial public acceptance of integrating the ChatGPT-type model with government services. *Kybernetes*. DOI: 10.1108/K-06-2023-1011.

Zhang N (2022) All-response" mechanism of government responsiveness: Case study of Beijing 12345 public service hotline. *Administrative Forum* 28: 78–86.

Zhang W, Yuan H, Zhu C, et al. (2022) Does citizen engagement with government social media accounts differ during the different stages of public health crises? An empirical examination of the COVID-19 pandemic. *Frontiers in Public Health* 10: 1–19. Available at: https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2022.807459/full (accessed 20 June 2024).

Zhipuqingyan (2023) ChatGLM2. Available at: https://chatglm.cn/ (accessed 20 June 2024).

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